RAMYA N B

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OBJECTIVE

To begin my career as a Design Verification Engineer by applying my knowledge of Digital Design, Verilog, SystemVerilog and UVM while enhancing my skills in the VLSI field.

EXPERIENCE

- **Trainee – VLSI GURU, Bangalore**
  APR 2025– Present
  - Trained in Digital Design, Verilog, SystemVerilog, UVM and AXI protocol
  - Designed and verified RTL digital designs
  - Developed reusable verification components
  - Performed simulation debugging
- **Graduate Trainee, Flight Test & Instrumentation (FTI), ARDC – Hindustan Aeronautics Limited (HAL), Bengaluru.**
  AUG 2024 – APR 2025

TECHNICAL SKILLS

Skills: Digital electronics, Verilog, SystemVerilog.
Verification Methodologies: UVM.
Scripting Language: Python
Operating systems: Linux
Tools: ModelSim, QuestaSim, EDA Playground.
Text Editors: GVIM

PROJECTS

1. AXI Master UVC Development – SystemVerilog & UVM

- Developed a simplified AXI Master UVC in SystemVerilog UVM supporting multiple burst types, burst lengths, and burst sizes. Created a behavioral AXI-slave responder to validate protocol-level interactions and data flow.
- Built Master Agent with sequencer, driver, and monitor to generate and sample AXI write/read transactions covering INCR, FIXED, and WRAP bursts with varying lengths and sizes
- Implemented a Responder (AXI Slave Model) to accept addresses, handle burst progression, compute wrap boundaries, store write data, and return read responses
- Developed a Scoreboard performing end-to-end checking using analysis ports: Expected transaction data from Driver Actual sampled data from Monitor

2. Memory Module–RTL Design, SystemVerilog & UVM based verification

- Designed a parameterized single-port memory module in Verilog with configurable address/data widths
- Integrated reset logic for robust initialization
- Developed and executed comprehensive test scenarios in SystemVerilog and UVM to validate memory functionality under different conditions
- Created reusable testbench components: driver, monitor and scoreboard
- Applied functional and code coverage with assertion.

3. Asynchronous FIFO–RTL Design, SystemVerilog & UVM based Verification

- Developed a parameterized asynchronous FIFO with separate read/write clocks
- Handled full, empty, overflow, and underflow conditions
- Ensured reset synchronization across clock domains
- Built a SystemVerilog and UVM testbench to validate functionality

EDUCATION

K V G College of Engineering Sullia, ECE(VTU) **2023** GRADE– 8.17 CGPA
Govt P U College Hosangadi **2019** SCORE:77.17%
Govt High School Hosangadi **2017** SCORE 84.32%

ACHIEVEMENTS

I was the Taluk-Level Athletics Champion, securing 1st place in the 800m, 1.5km, and 3km events. I also achieved 2nd and 3rd place in district-level athletics competitions and secured second place in the district-level Kho-Kho competition.

DECLARATION

I hereby declare that the details and information given above are complete and true to the best of my knowledge.

Signature: Ramya N B